

ONLINE MASTERS IN DATA SCIENCE

DSC 257R - UNSUPERVISED LEARNING

EXAMPLES OF NEURAL AUTOREGRESSIVE MODELS

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The Autoregressive Framework

To model d-dimensional data $X = (X_1, X_2, \dots, X_d)$:

- Pick an ordering of the variables, $\sigma(1), \dots, \sigma(d)$
- Explicitly learn the conditional probability functions

$$\Pr(x_{\sigma(i)} | x_{\sigma(1)}, \dots, x_{\sigma(i-1)})$$

For instance, we can fit neural nets to these functions.

NADE (Neural Autoregressive Density Estimation)

Larochelle, Murray (2011)

Binary variables:

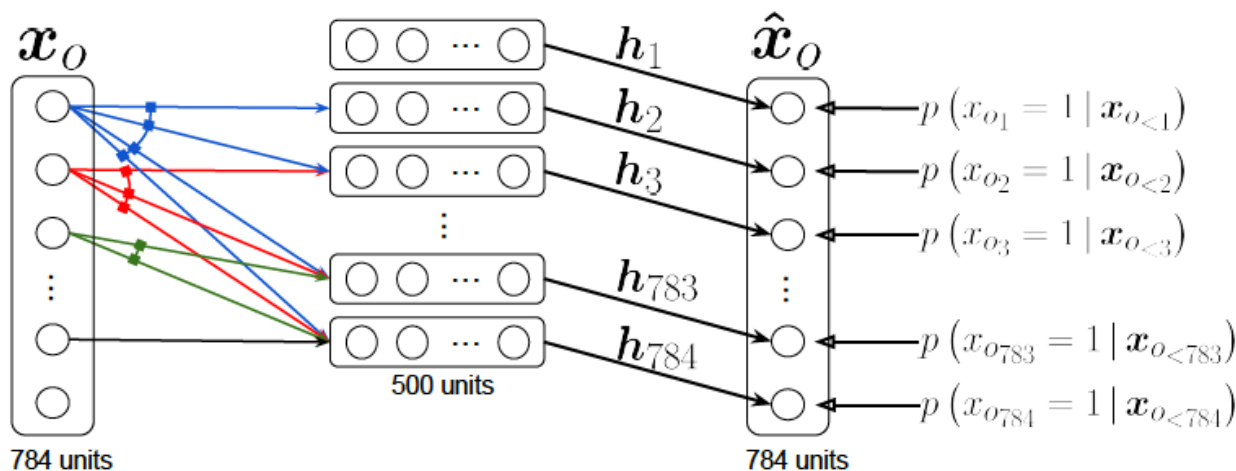
- Each $\Pr(x_{\sigma(i)} | x_{\sigma(1)}, \dots, x_{\sigma(i-1)})$ is a net with one hidden layer and sigmoid output.
- Tie weights to reduce the number of parameters.

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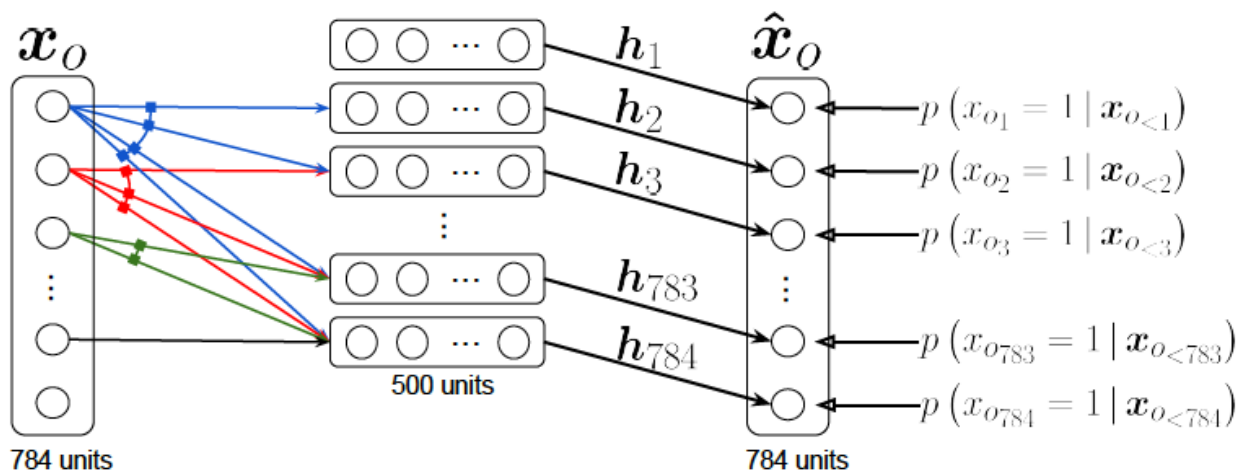


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Question: How to handle non-binary variables?

Handling General Variables

Need to specify $\Pr(x_{\sigma(i)} | x_{\sigma(1)}, \dots, x_{\sigma(i-1)})$

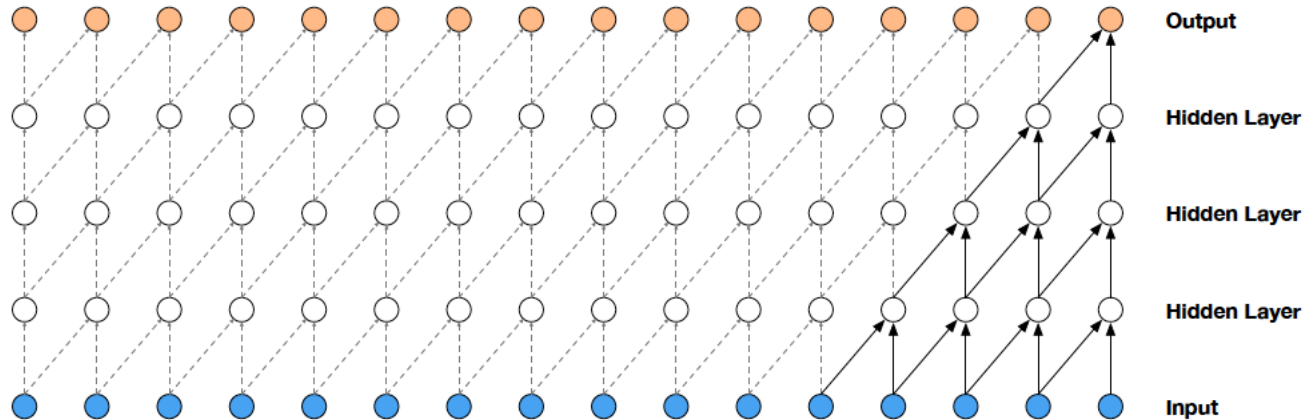
NADE on Binarized MNIST



Question: How can one evaluate a generative model like this?

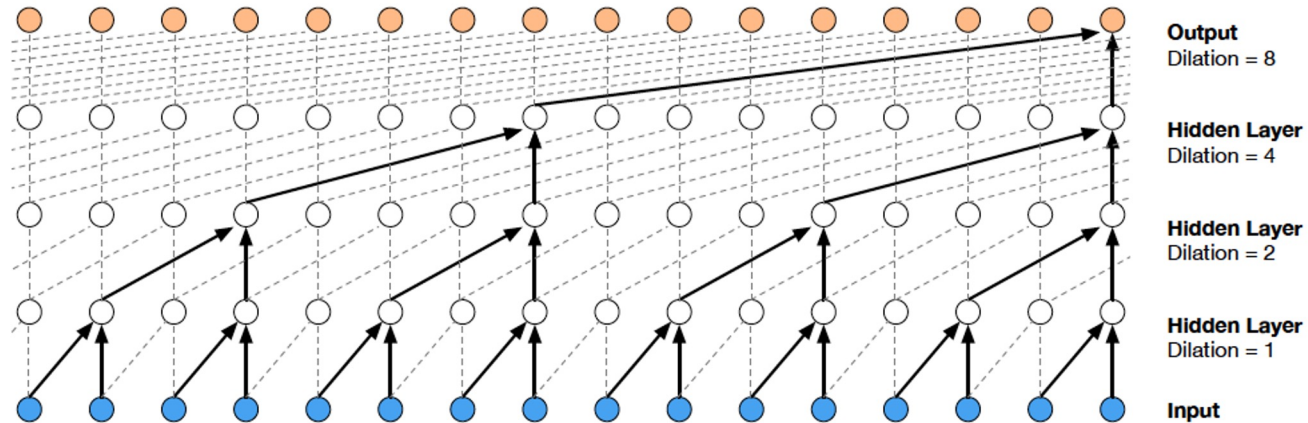
WaveNet

Van den Oord, Dieleman, Zen, Simoyan, Vinyals, Graves, Kalchbrenner, Senior, Kavukcuoglu (2016).



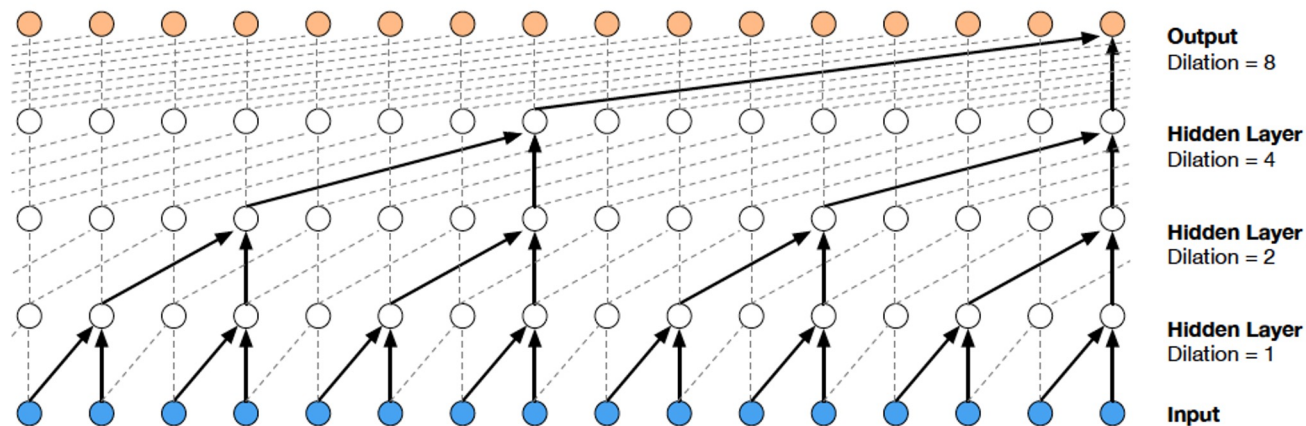
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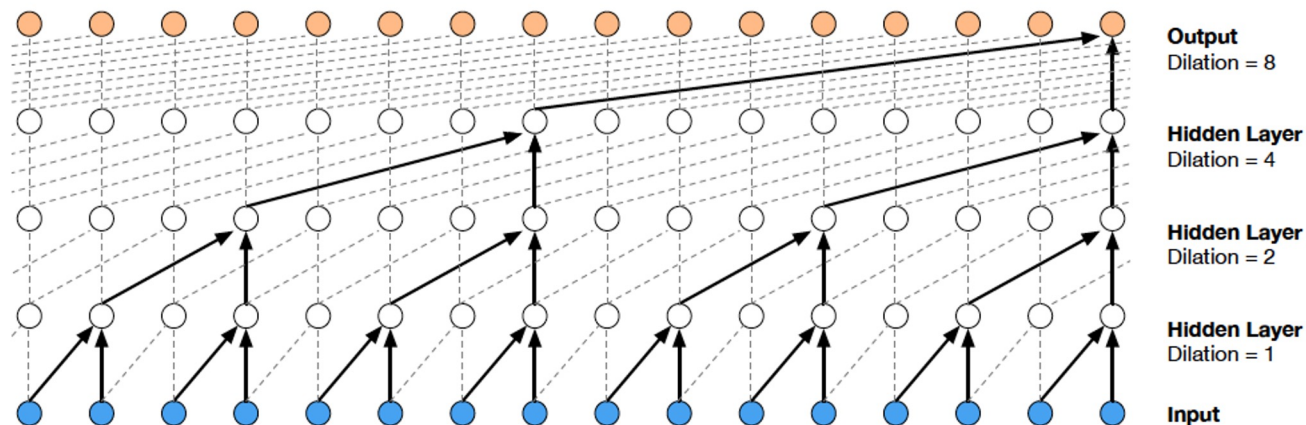
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- Output: Softmax over 256 values on a log-scale.

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- Output: Softmax over 256 values on a log-scale.
- Text-to-speech: conditional generation.